# Francesco **Mignone**

Turin

Open to relocation and remote work

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GitHub

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ilnerdchuck.github.io

**Programming Languages**

C
Assembly x86/RISC-V/ARM
SQL

**HDL Languages& Tools**

SV/VHDL/UVM
Modelsim/Questasim/Verdi
Design Compiler
Innovus
Quartus Prime
Z01X/VC-Z01X

**Scripting**

TCL
Bash/Shell

**Embedded Systems & Tools**

FreeRTOS
Arduino
Raspberry Pi
STM32

**Softwares**

Linux
Docker
KiCAD
GDB
Git

**Technical Skills**

Soldering
Logic Analyzers
PCB design
3D Printing & Design
LaTeX
Markdown

**Spoken Languages**

Italian - Native
English - C1 IELTS

For more information's on the projects or me visit my website!

## Education

2024 – March 2027
**Polytechnic University of Turin: Masters Degree in Embedded Systems**
Turin
**Coursework**: Microelectronics Systems, Synthesis and Optimization of Digital Systems, Electronics for Embedded Systems, Specification and Simulation of Digital Systems, Software Engineering, Cybersecurity For Embedded Systems, Operating Systems for Embedded Systems, Computer Architectures (RISC, ARM), Integrated System Architecture, Testing and Fault Tolerance, Electronic Systems Engineering
**Currently Remaining**: Energy Management for IoT, Embedded Architectures for AI and ML

2017 – 2024
**University of Pisa: Bachelor Degree in Computer Engineering**
Pisa
**Relevant Coursework**: Data Structures and Algorithms, Digital Logic Design, Computer Architectures (x86) and Kernel programming, Operating Systems, Electronic Circuits Analysis, Digital Electronics, Analog and Digital Communications, Computer Networks, Operations research, Automation Engineering
**Thesis**: Extension for VSCode to debug a multi-programmed Kernel

## Jobs and Internships

2026 – Today
**ARM - Internship and Master's Thesis: Functional Safety and Cybersecurity**
Sophia Antipolis
**VC-Z01X, SystemVerilog, Tcl, Scripting**
Dependability analysis of ARM architecture via fault injection.

## Projects

2025
**DLX Microprocessor**
**VHDL, Questasim, Design Compiler, Innovus**
RTL Design, Simulation, Synthesis and Physical Design of a DLX Microprocessor

- Pipeline implementation: Control Unit and Datapath
- Subset of DLX ISA: Load, Store, Arithmetic, Logic, Branch and Jump instructions
- Full ASIC design flow: RTL Design, Simulation, Synthesis and Physical Design
- Developed custom scripts to automate the Simulation, Synthesis and Physical Design process

2025
**NXP S32K on QEMU**
**C, QEMU, FreeRTOS**
Implementation of the NXP S32K3X8EVB board family in wemu with a set of emulated features:

- UART, SPI peripheral, TPM module for secure boot and secure key storage
- FreeRTOS porting for test and drivers for the peripherals

2025
**Verification Workshop**
**System Verilog, Questasim**
Hands-on sessions of SystemVerilog and UVM for hardware design and verification to test basic designs

2025
**Low-Power Contest**
**TCL, Prime Time, Design Compiler**
TCL script for post Synthesis analysis to optimize the leakage power of a design with a Multi-$V_{th}$ approach

2024 – Today
**GBA-SPi**
**KiCAD, RaspberryPi, Arduino**

- Schematic and PCB design
- LiPo battery protection and chagrin circuit
- External custom cartridge modules via GPIO
- ATMega power handling and I$^2$C communication with a RaspberryPi
- Custom RaspberryPi I$^2$C Driver for the ATMega